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Development of low latency, high resolution SMAP soil moisture retrievals in support of near-real-time applications

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Soil moisture plays a crucial role in weather forecasting and climate prediction, and provides useful information for flood and drought monitoring, and agricultural irrigation scheduling. In recent decades, soil moisture measurements have been enabled by space-borne passive microwave sensors. Nonetheless, the application of satellite remote sensing soil moisture data in operational meteorological and hydrological contexts is constrained by their prolonged latency and relatively coarse spatial resolution. This study describes the development of an operational enhanced Soil Moisture Active Passive (SMAP) soil moisture retrieval processor, referred to as SMAP_E_OPL, which addresses these limitations. SMAP_E_OPL produces near-real-time (NRT) soil moisture data at a 9-km grid resolution utilizing the single channel retrieval procedure employed in the long-latency SMAP Level 2 Enhanced soil moisture products (i.e., SMAP_L2SMP_E). In contrast to SMAP_L2SMP_E, the SMAP_E_OPL processor uses the Inverse Distance Squared Weighted interpolation approach to resample the NRT SMAP L1B brightness temperature (T_B) data from a 33-km resolution to the target grid for enhanced computational efficiency. Moreover, SMAP_E_OPL also allows the interoperable use of soil temperature and snow depth estimates from various operational land surface model (LSM) configurations. A bias-correction process is implemented to ensure consistency in the effective soil temperature (T_{eff}) between SMAP_E_OPL and SMAP_L2SMP_E. Evaluation based on *in situ* soil moisture measurements across the continental United States demonstrates that SMAP_E_OPL delivers the NRT soil moisture at 9-km grid resolution with accuracy comparable to the official (but high latency) SMAP enhanced soil moisture products. The use of different LSMs in SMAP_E_OPL to estimate T_{eff} yields similar quality in soil moisture retrievals, highlighting the flexibility of SMAP_E_OPL with respect to LSMs. These results confirm the viability of the SMAP_E_OPL processor in providing high-spatial-resolution soil moisture data with low latency for operational forecasting, data assimilation, and monitoring systems.

KEYWORDS
data assimilation, high resolution, near-real-time, soil moisture, Soil Moisture Active Passive (SMAP)

1 Introduction

Soil moisture plays a crucial role in weather forecasting and climate prediction systems by influencing the interactions between the atmosphere and Earth's surface (Entekhabi et al., 2010; Ochsner et al., 2013; Santanello Jr et al., 2019; Seneviratne et al., 2010). Timely and precise soil moisture data is essential for assessing natural disasters and managing water resources, including flood and drought monitoring (Crow et al., 2017; Engda and Kelleners, 2016; Massari et al., 2014; 2018; Mladenova et al., 2019) and agricultural irrigation scheduling (Moran et al., 2004). These factors are closely intertwined with the global economic stability, food security, and overall quality of life (Milman and Arsano, 2014). Nevertheless, capturing the spatial and temporal variability of soil moisture over large scales (e.g., continental or global) is a challenging endeavor, as ground-based measurements have limited coverage.

Satellite-based microwave observations at low frequencies (e.g., 1–10 GHz) can provide global estimates of near-surface soil moisture, given their properties of high sensitivity to soil moisture changes (Jackson et al., 2002; Njoku and Kong, 1977). The National Aeronautics and Space Administration (NASA) Soil Moisture Active Passive (SMAP) satellite, launched in January 2015, has been providing its standard near-surface soil moisture product since 31 March 2015 at a spatial resolution of 36 km with a revisit cycle of 2–3 days and a target accuracy of 0.04 m^3 m^{-3} with respect to the unbiased root-mean-square error (ubRMSE). Numerous studies (e.g., Bhuiyan et al., 2018; Chan et al., 2016; 2018; Chen et al., 2017; Colliander et al., 2017; Kumar et al., 2018; Pan et al., 2016; Wrona et al., 2017; Zhang et al., 2017) have demonstrated the capability of SMAP to monitor global soil moisture conditions. SMAP soil moisture retrievals have also been widely used in data assimilation (DA) systems to improve modeled soil moisture estimates (e.g., Ahmad et al., 2022; Blyverket et al., 2019; Chao et al., 2022; Kim et al., 2021; Kwon et al., 2022; Mladenova et al., 2019) and have been effectively applied to monitor droughts and floods (e.g., Chao et al., 2022; Mishra et al., 2017; Mladenova et al., 2019; Rahman et al., 2019; Sadri et al., 2018), assess cropland and food security (e.g., Gonsamo et al., 2019; Sadri et al., 2020), and characterize irrigation (e.g., Brocca et al., 2018; Kwon et al., 2022; Lawston et al., 2017; Nie et al., 2022).

Previous studies (e.g., Moran et al., 2004; Schmugge et al., 2002) have indicated that finer spatial resolution soil moisture is more effective in characterizing heterogeneous land surface hydrometeorological processes. Initially, SMAP was designed to provide high-resolution global measurements of near-surface soil moisture (at the top 5 cm soil depth) and freeze-thaw states with spatial resolution of up to 1–3 km, achieved by combining an L-band radar (active; 1.26 GHz) and an L-band radiometer (passive; 1.41 GHz) (Das et al., 2011; Entekhabi et al., 2010; Liu et al., 2016). However, the SMAP radar failed on 7 July 2015. Although radar observations from the European Space Agency's Sentinel-1 have been used to support a high-spatial resolution product at 1–3 km for the mission, known as SMAP/Sentinel-1 (SPL2SMAP_S, Das et al., 2019), the product has a long repeat coverage of around 12 days (Liu P.-W. et al., 2021). Various approaches have been developed to downscale the standard SMAP data, originally at 36 km resolution to finer grid resolutions, such as 9-km using optimal interpolation methods (Chan et al., 2018; O'Neill et al., 2019), and 1-km employing the thermal inertia algorithm (Fang et al., 2020; Liu Z. et al., 2021; Merlin et al., 2008) or by integrating thermal fluxes and geoinformation data (Liu et al., 2022).

In particular, soil moisture products with fine spatial resolution and low latency are a critical need for operational weather and hydrologic forecasting systems. For example, accurate soil moisture initial conditions are well-established as a significant requirement for numerical weather prediction (NWP) systems (Dirmeyer and Halder, 2016; Koster et al., 2010; Kwon et al., 2024; 2026). Having low latency products is, therefore, critical if those systems are to take advantage of information from satellite remote sensing. Such near-real-time (NRT) information is also essential for operational hydrological applications (Mladenova et al., 2019; Sadri et al., 2020) such as weather and flash flood forecasting. Rodríguez-Fernández et al. (2019) suggested that an ideal operational system needs to meet an NRT requirement of ~3 h, which is typically not met by current satellite mission products. For example, NASA's SMAP provides level 2 enhanced and standard soil moisture products at a latency of 14–16 h (nominal latency of 24 h). Its level 3 and 4 products require even longer time, ~2–3 days, for data processing and model run (Crow et al., 2017). Many processing steps are required for brightness temperature (T_B) observations and other ancillary data sources to generate soil moisture products from T_B using the geophysical retrieval algorithm. Particularly, an interpolation method, i.e., Backus-Gilbert (BG) algorithm (Poe, 1990), employed to produce the official SMAP Level 2 enhanced (9 km) soil moisture product (SMAP_L2SMP_E) usually takes 2–4 h to grid T_B onto the Equal-Area Scalable Earth (EASE)-Grid. It also relies on modeled soil temperature estimates from the NASA Global Modeling and Assimilation Office (GMAO) Goddard Earth Observing System-Forward Processing (GEOS-FP), which is not an operational environment. Thus, the process methodologies and reliance on external product sources may delay the data delivery of SMAP_L2SMP_E when extra processing and quality control procedures are included. On the other hand, the development of SMAP retrievals using ancillary information from the NASA Land Information System (LIS) modeling system results in the use of a consistent set of ancillary datasets between the remote sensing retrievals and NWP model.

A NRT version of the SMAP data (with ~3-h latency) has, in fact, been developed (O'Neill et al., 2022). However, this product only enables soil moisture data at the baseline coarse spatial resolution of 36 km. While such products have been assimilated in operational land DA systems for weather forecasting (Wegiel et al., 2020), the adoption of NRT SMAP products with finer spatial resolutions is hindered by their significant latency. Notably, Yin et al. (2020) introduced a downscaling method capable of generating NRT daily 1-km SMAP soil moisture data, yet this method relies on additional data sources, including the Visible Infrared Imaging Radiometer Suite (VIIRS) land surface temperature and the Moderate Resolution Imaging Spectroradiometer (MODIS) enhanced vegetation index, both of which are subject to cloud

cover limitations resulting in missing data (Yin et al., 2020). In this regard, the current study aims to develop an operational system that employs a more time-efficient gridding algorithm and uses system-self variables for the retrieval algorithm to fill these gaps.

This article describes the development of an operational, enhanced SMAP soil moisture retrieval, denoted as SMAP_E_OPL, with the capability to produce NRT hourly soil moisture products at a grid resolution ranging from 9 to 10 km, directly from the NRT SMAP Level 1B (L1B) T_B data (Piepmeier et al., 2020). This product is similar in grid resolution to the enhanced SMAP product, and it meets the NRT latency requirement of ~3 h. Here we demonstrate the following key enhancements: (a) The SMAP_E_OPL processor generates soil moisture information of comparable quality to the official (high latency) SMAP products, such as the SMAP Level 2 standard (36 km, SMAP_L2SMP) and enhanced Passive Soil Moisture Products (9 km, SMAP_L2SMP_E). (b) This is enabled by a flexible, open source implementation that utilizes inputs from different land surface models (LSMs) for ancillary information of effective soil temperature (T_{eff}) to yield similar performance in soil moisture retrievals. Note that both enhanced products (i.e., SMAP_L2SMP_E and SMAP_E_OPL) provide soil moisture data posted on a 9 km grid, while the effective spatial resolution remains approximately 33 km.

The SMAP_E_OPL soil moisture retrieval processor is integrated within the NASA LIS (Kumar et al., 2006; 2008) Framework (LISF), which is an open source software infrastructure for high performance land surface modeling and DA. LISF has a data processing environment called the Land Data Toolkit (LDT; Arsenault et al., 2018), which includes capabilities for processing land surface datasets for ingestion into the modeling environment LIS. The SMAP_E_OPL processor is integrated within the LDT subsystem of LISF. Several regional and global hydrological modeling and DA systems around the world have been enabled by LIS, including the North American land data assimilation system (NLDAS; Xia et al., 2012), the Global land data assimilation system (GLDAS; Rodell et al., 2004), the famine early warning land data assimilation system (FLDAS; McNally et al., 2017). Similarly, the U.S. Air Force (USAF) Weather also employs a custom configuration of LIS in their operations (Wegiel et al., 2020). We implemented the SMAP_E_OPL processor within this USAF environment to enable the real-time operational assimilation of the enhanced SMAP soil moisture data. This study validated the capability of the operational processor to generate high-spatial-resolution SMAP soil moisture retrievals in near-real-time, exhibiting accuracy comparable to official SMAP products. To the best of our knowledge, SMAP_E_OPL stands as the only operational processor that directly delivers NRT hourly enhanced SMAP soil moisture products (with grid resolution of 9 km) from the SMAP L1B T_B data. SMAP_E_OPL incorporates modeled T_{eff} and snow depth from LSMs within LIS, including 557th Weather Wing (WW) operational simulations, as well as other preprocessed static or climatological ancillary data.

TABLE 1 Main differences between the SMAP_E_OPL and SMAP_L2SMP_E soil moisture retrieval algorithms.

Feature	SMAP_E_OPL	SMAP_L2SMP_E
T_B data	SMAP NRT L1B T_B	SMAP standard (historical) L1B T_B
Interpolation method	Inverse distance squared weighted (IDSW) interpolation	Backus-Gilbert (BG) optimal interpolation
Spatial grid/Projection	0.14° (longitude) by 0.09° (latitude) equidistant cylindrical projection	9 km EASE grid 2.0 projection
Effective soil temperature (T_{eff})	Calculated from soil temperature simulated by LIS-LSMs	Calculated from soil temperature simulated by GMAO GEOS-FP
Snow flag	Based on snow depth from USAF-SI	Based on snow cover fraction from NOAA IMS
Frozen soil flag	Based on soil temperature from LIS-LSMs	Based on soil temperature from GMAO GEOS-FP

2 Materials and methods

2.1 Operational enhanced SMAP (SMAP_E_OPL) soil moisture retrieval processor

The SMAP_E_OPL soil moisture retrieval processor, integrated into LDT, follows the established protocol applied in the official SMAP L2 Enhanced soil moisture products (SMAP_L2SMP_E; O'Neill et al., 2019). However, certain modifications and adjustments are introduced to meet the NRT requirements for operational purposes. The primary distinctions include the source of the L1B T_B data, the interpolation method, spatial grid and projection, and ancillary data sources for parameters like T_{eff}, snow flag, and frozen soil flag, which are summarized in Table 1. It should be noted that while the objective of SMAP_E_OPL processor is to ingest the SMAP NRT T_B data, for the purpose of algorithm development and evaluation in this study, the standard SMAP L1B T_B data (Piepmeier et al., 2020) have been employed.

The SMAP_L2SMP_E processor utilizes the Backus-Gilbert (BG) optimal interpolation method (Poe, 1990) to resample the non-gridded SMAP Level 1B T_B to a 9-km EASE-Grid for the Level 1C Enhanced T_B Product (i.e., L1C_TB_E; Chaubell, 2016), requiring 2–4 h for resampling a half-orbit data. In contrast, the SMAP_E_OPL processor employs an Inverse Distance Squared Weighted (IDSW) interpolation method to resample the SMAP NRT L1B T_B data to a 0.14° longitude by 0.09° latitude target grid using the equidistant cylindrical projection. This grid configuration is

used in SMAP_E_OPL to be consistent with the global modeling configuration used at 557th WW. The IDSW interpolation is a conventional technique utilized in passive microwave satellite missions like the Advanced Microwave Scanning Radiometer for Earth Observing System (AMSR-E) (Knowles et al., 2006) and reduces the time for T_B gridding to approximately 3 min per half-orbit. Despite the efficiency gain, the SMAP_E_OPL processor maintains comparable skill to the standard processor, with just around 1 K difference in T_B resulting in ~0.01 m^3 m^{-3} difference in soil moisture.

The resampled T_B data (SMAP NRT L1C) is utilized in soil moisture retrieval algorithms, incorporating T_{eff} from LIS-LSMs as detailed in Section 2.3, along with ancillary data to generate the SMAP_E_OPL soil moisture. This involves employing a traditional τ (vegetation optical depth)-ω (albedo scattering) model (Equation 1) with the Mironov model (Equations 2–10) to convert the dielectric constant into soil moisture, consistent with the algorithm used for the SMAP standard soil moisture products.

$$T_{B,p} = T_{eff} e_p \exp\left(-\tau_p \sec\theta_l\right) + T_{vege}\left(1-\omega_p\right)\left[1-\exp\left(-\tau_p \sec\theta_l\right)\right]\left[1+r_p \exp\left(-\tau_p \sec\theta_l\right)\right] \quad (1)$$

where e_p is the emissivity, θ_l the SMAP look angle (=40°), T_{vege} is the vegetation canopy temperature, and r_p is the reflectivity. The key regression equations used in Mironov's generalized refractive mixing dielectric model (GRMDM; Mironov et al., 2009) are shown below:

$$n_d = 1.634 - 0.539\times10^{-2}C + 0.2748\times10^{-4}C^2 \quad (2)$$

$$k_d = 0.03952 - 0.04038\times10^{-2}C \quad (3)$$

$$m_{vt} = 0.02863 + 0.30673\times10^{-2}C \quad (4)$$

$$\varepsilon_{0b} = 79.8 - 85.4\times10^{-2}C + 32.7\times10^{-4}C^2 \quad (5)$$

$$\tau_b = 1.062\times10^{-11} + 3.450\times10^{-12}\times10^{-2}C \quad (6)$$

$$\sigma_b = 0.3112 + 0.467\times10^{-2}C \quad (7)$$

$$\sigma_u = 0.3631 + 1.217\times10^{-2}C \quad (8)$$

$$\varepsilon_{0u} = 100 \quad (9)$$

$$\tau_u = 8.5\times10^{-12} \quad (10)$$

where n_d is the refractive index of dry soil; k_d is the normalized attenuation coefficient of dry soil; m_{vt} is the maximum bound water fraction; ε_{0b} and ε_{0u} denote the low-frequency limits of the dielectric constants for bound and free soil water, respectively; τ_b and τ_u are the relaxation times for bound and free soil water, respectively; σ_b and σ_u represent the relaxation time for bound and free soil water, respectively; and C is the clay content. For more details on the τ-ω and Mironov models, refer to the SMAP Algorithm Theoretical Basis Document (ATBD; O'Neill et al., 2021a) and Mironov et al. (2009). Within SMAP_E_OPL, the SMAP Single Channel Algorithm using vertically polarized T_B (SCA-V) is employed. We selected SCA-V for the current version of SMAP_E_OPL because it has served as a baseline algorithm for the SMAP product and has been extensively evaluated and used in many previous studies. Nevertheless, we note that the dual-channel algorithm (DCA) has recently been adopted as the SMAP baseline product and that, in some land cover types, SCA using horizontally polarized T_B (SCA-H) provides better quality (Judge et al., 2021). In this study, however, we use SCA-V to generate the SMAP_E_OPL soil moisture data and to compare them with the official SMAP data, which are also produced using SCA-V. We consider this choice reasonable for the processor-development objectives of this study. The incorporation of DCA will be considered in future versions of the SMAP_E_OPL processor. Figure 1 presents a schematic diagram of SMAP_E_OPL. The snow flag and frozen soil flag are obtained from the operational United States (US) Air Force Snow and Ice Analysis (USAF-SI; Yoon et al., 2022) and the NRT US Air Force 557th WW global operational LIS-LSMs.

T_{eff} serves as a crucial factor in the τ-ω model (O'Neill et al., 2021a). In the SMAP_E_OPL processor, T_{eff} is computed using soil temperature data from LIS-LSMs at the same time of the previous day (t-1 day), which is the identical setting as used in the official SMAP NRT SM product, while the SMAP_L2SMP_E processor relies on soil temperature simulated by the GEOS-FP system from NASA GMAO at t. The utilization of the LIS-based soil temperature from the previous day (t-1 day) reduces the reliance of the retrieval processor on data sources that are not available in real-time, thereby mitigating potential delays in soil moisture retrievals. However, it should be noted that the use of t-1 day soil temperature estimates may introduce errors into the SMAP_E_OPL soil moisture retrievals. LIS-LSMs utilized in this study encompass the Noah LSM (Ek et al., 2003) version 3.9, Noah LSM with multi-parameterization options (Noah-MP; Niu et al., 2011; Yang et al., 2011) version 4.0.1, and Joint United Kingdom (UK) Land Environment Simulator (JULES; Best et al., 2011; Clark et al., 2011) version 5.0. To ensure consistency between the SMAP_E_OPL and SMAP_L2SMP_E soil moisture products, the T_{eff} derived from the NRT US Air Force 557th WW global operational LIS-LSMs undergoes bias-correction in SMAP_E_OPL, as outlined in Section 2.3.

The snow flag within SMAP_E_OPL relies on snow depth simulated by USAF-SI at t-3h while SMAP_L2SMP_E employs snow cover fraction data from the National Oceanic and Atmospheric Administration (NOAA) Interactive Multisensor Snow and Ice Mapping System (IMS) for the snow flag. Additionally, the frozen soil flag in SMAP_E_OPL is determined based on soil temperature estimates from the NRT US Air Force 557th WW global operational LIS-LSMs, while the GMAO GEOS-FP soil temperature outputs are utilized in SMAP_L2SMP_E. These flags are applied to exclude soil moisture retrievals for pixels covered by over 5 cm of snow or categorized as frozen soil. The difference in the snow and frozen soil flag sources between SMAP_E_OPL and SMAP_L2SMP_E may cause some disagreement between them for missing soil moisture data points.

The retrieval algorithms in SMAP_E_OPL require additional ancillary data, including land cover, soil texture (i.e., clay fraction and bulk density), surface geometry (roughness), and vegetation information (τ and ω) (O'Neill et al., 2021a). These

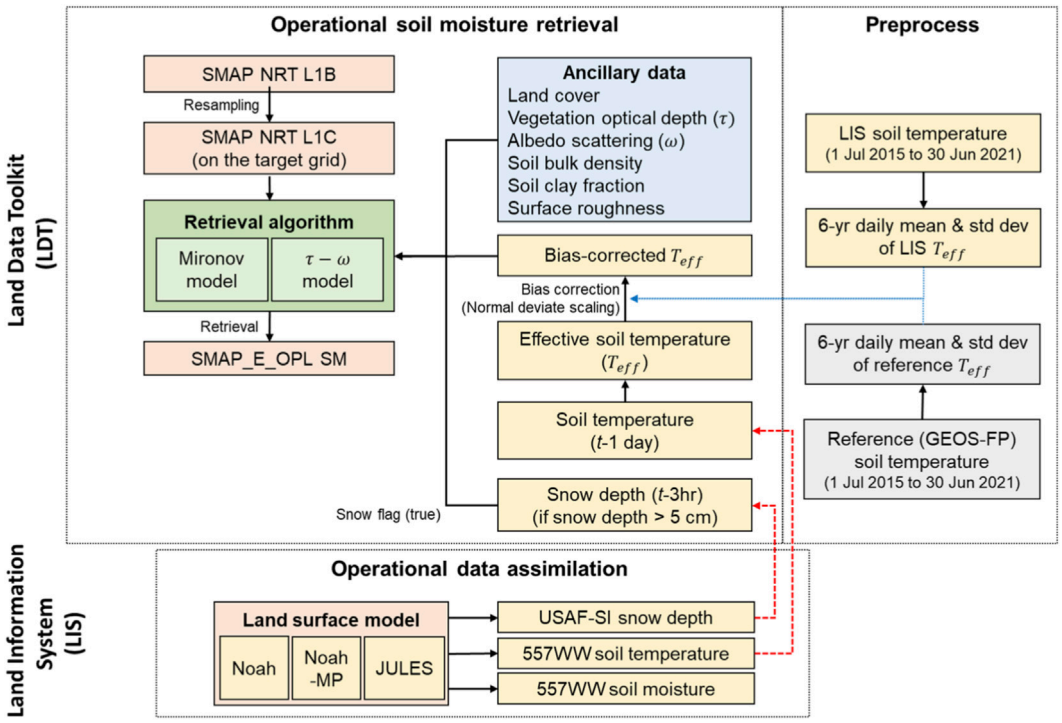


FIGURE 1
Schematic diagram of the operational enhanced SMAP (SMAP_E_OPL) soil moisture retrieval processor in the Land Data Toolkit (LDT) that is linked to the operational soil moisture data assimilation system in the Land Information System (LIS).

datasets in SMAP_E_OPL match those utilized for the SMAP_L2SMP_E soil moisture products but are regridded to the target grid (0.14° longitude by 0.09° latitude, equidistant cylindrical projection). Clay fraction and soil bulk density are sourced from the global SoilGrid250m (Hengl et al., 2017) (available at https://www.isric.org/explore/soilgrids), aggregated by averaging values to the target grid. The surface roughness and ω values are determined from a lookup table, based on the primary land cover type in a grid, derived from the Terra MODIS-based International Geosphere-Biosphere Programme (IGBP) land cover classes (Friedl et al., 2002) at their original 500 m spatial resolution, selecting the most representative class (i.e., the highest percentage of presence class) within the target grid. The ω values used in this study range from 0.03 to 0.08 (Supplementary Table 1). τ (ranging from ~0.2 to 1.0) is estimated from vegetation water content (VWC) multiplied by an empirical parameter (b) acquired from a lookup table for each land cover class (Supplementary Table 1). VWC is determined from the MODIS Normalized Difference Vegetation Index (NDVI) climatology from 2000 to 2010, consistent with the data used for SMAP_L2SMP_E. Further details of these processes are outlined in the SMAP ATBD (O'Neill et al., 2021a).

2.2 Land surface model setup

Here we utilize T_{eff} estimates from three different LIS-LSMs including Noah, Noah-MP, and JULES within the SMAP_E_OPL processor. These three LSMs were selected because they are the operational models implemented in the US Air Force 557th WW, the forecast system targeted in this study. However, our results (see Section 3.1) highlight the flexibility of the SMAP_E_OPL processor, which provides consistent soil moisture retrieval performance across different LSMs.

The Noah LSM is based on the Oregon State University (OSU) LSM and has been extensively used in coupled numerical weather and climate prediction models due to its computational efficiency (Wei et al., 2013). Noah simulates soil and land surface temperature, canopy water content, soil moisture, snow water equivalent, and water and energy fluxes. The vegetation seasonality is represented by the spatially and temporally varying (monthly) green vegetation cover fraction. In the Noah LSM, a grid cell is composed of a vegetation canopy layer, a single snow layer, and four soil layers with thicknesses of 0.1 m, 0.3 m, 0.6 m, and 1.0 m from top to bottom (with a total soil depth of 2 m), for which soil moisture and temperature are estimated using the diffusivity form of the Richard's equation and the one-dimensional thermal diffusion, respectively. Noah computes evapotranspiration as the sum of vegetation transpiration, evaporation from the soil surface, and evaporation from canopy-intercepted water, and it calculates the latent heat flux based on the Penman potential evaporation approach. Vegetation rooting depth is estimated for each vegetation type by applying a seasonal factor suggested by Dickinson et al. (1993) to represent the seasonality of transpiration and soil moisture (Wei et al., 2013). The subgrid-scale variability of soil moisture and precipitation is modeled by the surface moisture infiltration scheme based on Schaake et al. (1996), and moisture

infiltration during the winter season is influenced by the presence of frozen soil.

Compared to the baseline Noah LSM, Noah-MP provides multiple enhanced representations of hydrological processes related to vegetation, soil, snow, and groundwater (Niu et al., 2011; Yang et al., 2011). Noah-MP represents a one-dimensional column within each model grid cell, consisting of a vegetation canopy layer, up to three snowpack layers (depending on snow depth), a four-layer soil with the same default thicknesses as in the Noah LSM, and an unconfined aquifer layer. In Noah-MP, a semi-tile scheme is used to account for subgrid-scale heterogeneity (Niu et al., 2011). Unlike other LSMs, Noah-MP offers multiple parameterization options, enabling multi-physics ensemble simulations with diverse physical scheme combinations in a single model (Cai et al., 2014). This study employs the following physical scheme options in Noah-MP: static vegetation option for leaf area index and green vegetation fraction, using monthly prescribed values from input datasets; the Ball–Berry scheme for canopy stomatal resistance (Ball et al., 1987); the Noah-type soil moisture factor for stomatal resistance (Chen and Dudhia, 2001); the Monin–Obukhov similarity scheme (Brutsaert, 1982) for surface layer drag coefficient calculation; the TOPography-based hydrological MODEL (TOPMODEL) runoff with the simple groundwater scheme (SIMGM) (Niu et al., 2007); the modified two-stream scheme for radiative transfer (Niu and Yang, 2004; Yang and Friedl, 2003); the NY06 method (Niu and Yang, 2006) for representing supercooled liquid water and soil permeability under frozen conditions; the Biosphere-Atmosphere Transfer Scheme (BATS) parameterization for snow albedo (Yang and Dickinson, 1996); the semi-implicit time schemes for snow and soil temperature; the original Noah approach for parameterizing the lower-boundary conditions of soil temperature; the Jordan scheme (Jordan, 1991) for partitioning precipitation into snow and rain.

Like Noah and Noah-MP, JULES is a process-based community LSM, developed from its baseline predecessor, the Met Office Surface Exchange Scheme (MOSES; Cox et al., 1999), which simulates land-atmosphere interactions. JULES represents subgrid heterogeneity of the land surface using subgrid tiles, which include nine land cover classes: broadleaf trees, needleleaf trees, C3 (temperate) grass, C4 (tropical) grass, shrubs, bare soil, land ice, inland water, and urban areas. In JULES, soil moisture and temperature are modeled in four layers with thicknesses of 0.1 m, 0.25 m, 0.65 m, and 2.0 m from the surface to the bottom, extending to a total soil depth of 3 m. JULES has single- and multi-layer snowpack options, the latter of which is employed in this study. The model computes the surface energy fluxes for each land cover type within a grid cell, which are then averaged using weights. Precipitation reaching the ground after canopy interception is partitioned between surface runoff and infiltration. The vertical redistribution of infiltrated water between soil layers is represented using a finite-difference approximation of the Richards equation (Richards, 1931). JULES estimates soil hydraulic conductivity using the van Genuchten (van Genuchten, 1980) formulation, accounting for the effects of frozen soil.

These LSMs are driven by the meteorological forcing generated in operational settings at the 557th WW at the United States Air Force (Kemp et al., 2022). The forcing includes precipitation, air temperature, solar radiation, atmospheric pressure, and wind speed from the United States Air Force Global Air-Land Weather Exploitation Model (GALWEM), NOAA Global Forecast System (GFS), NASA Integrated Multiscale Retrievals for Global Precipitation Measurement (IMERG; Huffman, et al., 2020), World-Wide Merged Cloud Analysis (Air Force Weather Agency, 2013), along with *in situ* surface meteorology observations (Kemp et al., 2022). Land surface geophysical parameters including static land use/cover, soil texture, and topography, and monthly climatological albedo and greenness are summarized in Supplementary Table 2. The LIS-LSMs operate globally at a spatial resolution of 0.14° (longitude) by 0.09° (latitude) delivering 3-hourly instantaneous soil temperature and 6-hourly USAF-SI snow depth outputs. These outputs serve as inputs for the SMAP_E_OPL processor to generate soil moisture retrievals covering the period from 1 April 2015 to 31 December 2021. Note that soil moisture is retrieved whenever T_B measurements are available, whereas 3-hourly soil temperature outputs from LIS-LSMs are used in the retrieval processor due to the design of the US Air Force 557th WW system. Throughout this paper, the SMAP_E_OPL retrievals employing Noah, Noah-MP, and JULES are respectively referred to as SMAP_E_OPL (Noah), SMAP_E_OPL (Noah-MP), and SMAP_E_OPL (JULES).

2.3 Bias correction of effective soil temperature (T_{eff})

In SMAP_L2SMP_E, T_{eff} is calculated using Equation 11, modified from the Choudhury's two-layer T_{eff} model (Choudhury et al., 1982).

$$T_{eff} = K \cdot \left[C \cdot T_{s1} + (1 - C) \cdot T_{s2} \right] \tag{11}$$

where T_{s1} and T_{s2} are the average temperature of the first and second soil layers, respectively, of the GMAO GEOS-FP model. The empirical parameters, K (=1.007) and C [=0.246 for AM (descending overpasses) and 1.000 for PM (ascending overpasses) soil moisture retrievals] are obtained from extensive calibration processes using the GEOS-FP soil temperature and *in situ* temperature measurements for SMAP_L2SMP_E (O'Neill et al., 2021a).

However, systematic differences may exist between the soil temperature obtained from the LIS-LSMs (Noah, Noah-MP, and JULES used in SMAP_E_OPL) and that simulated by GMAO GEOS-FP (utilized in SMAP_L2SMP_E), owing to differing model physics and representation of vertical soil layers. Therefore, in order to ensure consistency between SMAP_E_OPL and SMAP_L2SMP_E, the SMAP_E_OPL processor employs a bias-correction approach. In this approach, SMAP_E_OPL adopts the same K and C values as used in SMAP_L2SMP_E, and rescales the LIS-LSM-based T_{eff} to the climatology of the GEOS-FP T_{eff} through the normal deviate scaling method (Equation 12), as follows:

$$T'_{eff_lis} = \left(T_{eff_lis} - \overline{T_{eff_lis}} \right) \cdot \frac{\sigma_{ref}}{\sigma_{lis}} + \overline{T_{eff_ref}} \tag{12}$$

where T_{eff_lis} and T'_{eff_lis} are the LIS-LSM-based T_{eff} at local 6 a.m. or 6 p.m. before and after bias-correction, respectively; $\overline{T_{eff_lis}}$ and σ_{lis} are the preprocessed 6-year (2015–2021) daily mean and standard deviation, respectively, of the LIS-LSM T_{eff} at local 6 a.m. or 6 p.m.; and $\overline{T_{eff_ref}}$ and σ_{ref} are the preprocessed 6-year (2015–2021)

daily mean and standard deviation, respectively, of the reference model (i.e., GEOS-FP) T_{eff} at local 6 a.m. or 6 p.m. The scaling approach increases the flexibility of SMAP_E_OPL with respect to using different LSMs. The use of the normal deviate scaling method, rather than a more refined approach such as cumulative distribution function (CDF) matching, is based on its relatively stable performance with limited sample sizes.

The spatial resolution of the GMAO GEOS-FP's soil temperature used in the official 9-km SMAP product (i.e., SMAP_L2SMP_E) is 0.3° (longitude) × 0.25° (latitude) while that of the LIS-LSMs' soil temperature used in this study is 0.14° (longitude) × 0.09° (latitude). LDT reads the GEOS-FP soil temperature of the top two soil layers and resamples it to the target grid (i.e., 0.14° (longitude) × 0.09° (latitude) in this study) using the bilinear interpolation method, the same approach used for the SMAP official products. The interpolated soil temperatures are then used to calculate T_{eff}. The difference in the spatial resolution between GEOS-FP and LIS-LSMs (and the resampling procedure employed in LDT) may be one of the potential sources causing the difference in the retrieved soil moisture between SMAP_E_OPL and SMAP_L2SMP_E.

Note that the primary purpose of the current study is to develop an operational retrieval processor (a) that generates enhanced (9-km) SMAP soil moisture data whose quality is comparable to that of the official SMAP SM product (i.e., SMAP_L2SMP_E) while achieving reduced latency suitable for operational systems that are sensitive to data latency, and (b) that are applicable to any operational systems. For this reason, we adopted a bias correction procedure (i.e., normal deviate scaling), which increases the flexibility of the processor for different systems as compared to directly using soil temperature from LIS-LSMs by recalibrating the empirical parameters (i.e., K and C) in Equation 11.

2.4 *In situ* soil moisture measurements

The performance of SMAP_E_OPL is evaluated over the continental United States (CONUS) by comparing to *in situ* soil moisture measurements from the United States Department of Agriculture (USDA) Soil Climate Analysis Network (SCAN, https://www.wcc.nrcs.usda.gov/scan; Diamond et al., 2013; Schaefer et al., 2007) and United States Climate Reference Network (USCRN, https://www.ncei.noaa.gov/access/crn; Bell et al., 2013). These sparse networks consist of monitoring stations distributed across diverse land cover classes and hydroclimate zones and have been used for assessing various SMAP products (O'Neill et al., 2021b). SCAN and USCRN stations measure hourly soil moisture at five soil depths of 5, 10, 20, 50, and 100 cm using a single set of probes and triplicate soil moisture probes, respectively (Bell et al., 2013; Diamond et al., 2013). Evaluation involves the station-to-pixel comparison of the soil moisture retrievals from SMAP_E_OPL, SMAP_L2SMP_E, and SMAP_L2SMP against the measured soil moisture at a 5-cm soil depth following the Sparse Network Assessment used in the official SMAP quality assessment report (O'Neill et al., 2021b). A total of 153 stations (SCAN: 97 stations; USCRN: 56 stations) are employed, selected based on identical quality-control protocols as used in Chen et al. (2018).

3 Results

We first focus on demonstrating whether the high-spatial-resolution NRT soil moisture products from the SMAP_E_OPL processor implemented in LDT exhibit comparable quality to those from the official, high latency SMAP_L2SMP_E processor. In this regard, soil moisture retrievals from SMAP_E_OPL, SMAP_L2SMP_E, and SMAP_L2SMP are evaluated using *in situ* soil moisture measurements over CONUS. In the second subsection, we delve into the application of the soil moisture data generated in this study within the context of DA techniques. We highlight how this integration is poised to enhance hydrological predictions and contribute significantly to water resources management applications.

3.1 SMAP_E_OPL soil moisture processor: a comparative analysis using SCAN and USCRN *in situ* data

The performance of the soil moisture retrievals from the SMAP_E_OPL processor is evaluated using *in situ* soil moisture data from the SCAN and USCRN stations for the period from 2015 to 2021. The sparse network assessment is a station-to-pixel evaluation, identical to the methodology used in the SMAP assessment report (O'Neill et al., 2021b). The results are summarized in Table 2. The SMAP_E_OPL processor utilizes three different sources of effective soil temperature (T_{eff}) from the Noah, Noah-MP, and JULES models to assess algorithm stability. Additionally, soil moisture products from SMAP_L2SMP_E and SMAP_L2SMP are used to compare the quality of SMAP_E_OPL with the mission products. Note must be taken that the spatial resolution of the SMAP_L2SMP_E product is 33 km posted on a 9-km EASE-GRID, and that of the SMAP_L2SMP product is 36 km posted on a 36-km EASE-GRID; while the soil moisture retrieval from SMAP_E_OPL is at 33 km posted on a 0.14° (longitude) × 0.09° (latitude) grid, which is at a similar resolution as SMAP products. The objective of this assessment is to demonstrate that the soil moisture from SMAP_E_OPL and SMAP_L2SMP_E is of comparable skill.

The evaluation employs statistical metrics such as unbiased root mean square error (ubRMSE), RMSE, bias, and correlation coefficient (R) for descending (AM), ascending (PM), and combined overpasses (AM + PM) over CONUS. Note that while RMSE and bias are included as metrics to compare different soil moisture retrievals, they are influenced by systematic bias and representativeness errors inherent in station-based evaluations. For all stations used in this study, the domain-averaged ubRMSE, RMSE, and R values for SMAP_E_OPL soil moisture retrievals during AM + PM overpasses are approximately 0.052 $m^3\ m^{-3}$, 0.074 $m^3\ m^{-3}$, and 0.65, respectively. These values are comparable to those of SMAP_L2SMP_E (ubRMSE 0.050 $m^3\ m^{-3}$, RMSE 0.073 $m^3\ m^{-3}$, and R 0.68) and SMAP_L2SMP (ubRMSE 0.050 $m^3\ m^{-3}$, RMSE 0.074 $m^3\ m^{-3}$, and R 0.67). Comparable performance is also obtained when soil moisture retrievals are evaluated separately for the AM and PM overpasses. Furthermore, all retrieval processors slightly overestimate soil moisture (0 < domain-averaged bias ≤ 0.007 $m^3\ m^{-3}$). Although the performance of SMAP_E_OPL is marginally worse compared to the official SMAP soil moisture products (SMAP_L2SMP_E and SMAP_L2SMP), partly because

TABLE 2 Domain-averaged evaluation metrics (± standard deviation) of soil moisture retrievals from SMAP_E_OPL using three LSMs (i.e., Noah, Noah-MP, and JULES), SMAP_L2SMP_E, and SMAP_L2SMP over CONUS. The metrics are computed by comparing each soil moisture retrieval against the *in situ* soil moisture measurements (i.e., SCAN and USCRN) from 1 April 2015 to 31 December 2021.

Product	ubRMSE (m^3 m^{-3})			RMSE (m^3 m^{-3})			Bias (m^3 m^{-3})			R		
	AM + PM	AM	PM	AM + PM	AM	PM	AM + PM	AM	PM	AM + PM	AM	PM
SCAN + USCRN												
SMAP_E_OPL (Noah)	0.052 (±0.016)	0.052 (±0.016)	0.052 (±0.017)	0.074 (±0.033)	0.073 (±0.033)	0.075 (±0.034)	0.004 (±0.061)	0.005 (±0.059)	0.003 (±0.062)	0.646 (±0.143)	0.657 (±0.132)	0.639 (±0.161)
SMAP_E_OPL (Noah-MP)	0.052 (±0.017)	0.051 (±0.017)	0.052 (±0.017)	0.075 (±0.033)	0.073 (±0.033)	0.076 (±0.035)	0.006 (±0.061)	0.005 (±0.059)	0.007 (±0.063)	0.646 (±0.140)	0.658 (±0.131)	0.634 (±0.160)
SMAP_E_OPL (JULES)	0.052 (±0.016)	0.052 (±0.016)	0.052 (±0.017)	0.074 (±0.033)	0.073 (±0.033)	0.075 (±0.034)	0.003 (±0.060)	0.004 (±0.059)	0.002 (±0.062)	0.651 (±0.138)	0.655 (±0.132)	0.649 (±0.149)
SMAP_L2SMP_E	0.050 (±0.017)	0.050 (±0.017)	0.050 (±0.017)	0.073 (±0.033)	0.072 (±0.033)	0.073 (±0.035)	0.003 (±0.061)	0.005 (±0.059)	0.002 (±0.062)	0.675 (±0.136)	0.679 (±0.129)	0.675 (±0.146)
SMAP_L2SMP	0.050 (±0.017)	0.050 (±0.017)	0.050 (±0.017)	0.074 (±0.033)	0.073 (±0.033)	0.074 (±0.034)	0.005 (±0.061)	0.007 (±0.059)	0.004 (±0.062)	0.669 (±0.137)	0.673 (±0.131)	0.669 (±0.148)
SCAN												
SMAP_E_OPL (Noah)	0.052 (±0.017)	0.052 (±0.017)	0.052 (±0.017)	0.071 (±0.027)	0.069 (±0.027)	0.071 (±0.028)	0.006 (±0.053)	0.007 (±0.051)	0.004 (±0.054)	0.627 (±0.153)	0.640 (±0.142)	0.616 (±0.173)
SMAP_E_OPL (Noah-MP)	0.052 (±0.017)	0.051 (±0.017)	0.052 (±0.017)	0.071 (±0.027)	0.069 (±0.027)	0.072 (±0.028)	0.007 (±0.053)	0.007 (±0.051)	0.008 (±0.054)	0.626 (±0.151)	0.640 (±0.141)	0.612 (±0.172)
SMAP_E_OPL (JULES)	0.052 (±0.017)	0.052 (±0.017)	0.052 (±0.017)	0.071 (±0.027)	0.070 (±0.027)	0.071 (±0.028)	0.005 (±0.053)	0.006 (±0.051)	0.004 (±0.054)	0.632 (±0.147)	0.637 (±0.141)	0.628 (±0.159)
SMAP_L2SMP_E	0.050 (±0.017)	0.050 (±0.017)	0.050 (±0.017)	0.070 (±0.026)	0.068 (±0.026)	0.069 (±0.027)	0.004 (±0.052)	0.007 (±0.050)	0.002 (±0.052)	0.655 (±0.145)	0.661 (±0.138)	0.653 (±0.156)
SMAP_L2SMP	0.050 (±0.017)	0.050 (±0.017)	0.050 (±0.017)	0.070 (±0.028)	0.069 (±0.028)	0.070 (±0.029)	0.006 (±0.053)	0.008 (±0.052)	0.005 (±0.054)	0.649 (±0.148)	0.655 (±0.141)	0.646 (±0.160)
USCRN												
SMAP_E_OPL (Noah)	0.052 (±0.016)	0.052 (±0.017)	0.051 (±0.016)	0.081 (±0.041)	0.080 (±0.040)	0.081 (±0.043)	0.001 (±0.073)	0.001 (±0.071)	0.001 (±0.075)	0.680 (±0.117)	0.686 (±0.107)	0.677 (±0.132)
SMAP_E_OPL (Noah-MP)	0.052 (±0.016)	0.052 (±0.017)	0.052 (±0.016)	0.081 (±0.041)	0.079 (±0.040)	0.082 (±0.044)	0.003 (±0.073)	0.002 (±0.071)	0.005 (±0.076)	0.679 (±0.114)	0.689 (±0.106)	0.673 (±0.129)
SMAP_E_OPL (JULES)	0.052 (±0.016)	0.052 (±0.016)	0.051 (±0.016)	0.080 (±0.041)	0.079 (±0.040)	0.081 (±0.043)	0.000 (±0.072)	0.001 (±0.071)	−0.000 (±0.075)	0.684 (±0.115)	0.687 (±0.107)	0.684 (±0.123)
SMAP_L2SMP_E	0.050 (±0.017)	0.050 (±0.017)	0.049 (±0.016)	0.080 (±0.042)	0.080 (±0.041)	0.081 (±0.044)	0.001 (±0.074)	0.003 (±0.072)	0.000 (±0.076)	0.709 (±0.111)	0.709 (±0.106)	0.713 (±0.118)
SMAP_L2SMP	0.051 (±0.017)	0.051 (±0.018)	0.050 (±0.017)	0.080 (±0.041)	0.079 (±0.040)	0.080 (±0.042)	0.003 (±0.072)	0.005 (±0.071)	0.001 (±0.074)	0.703 (±0.110)	0.702 (±0.107)	0.708 (±0.117)

it uses previous-day soil temperature estimates rather than the current-day estimates used by the official non-NRT products, the difference is less than the uncertainty of conventional ground-based soil moisture sensors. It is essential to emphasize that SMAP_E_OPL can provide timely high-spatial-resolution soil moisture information whose latency is determined solely by that of the SMAP L1B T_B data, and thus is suited for operational forecasting and monitoring systems. Soil moisture retrievals from the descending (AM) orbit in general show lower ubRMSE, lower RMSE, and higher R than those from the ascending (PM) orbit, although some exceptions exist.

Additionally, it is evident from Table 2 that the use of different LSMs does not lead to significant differences in the domain-averaged performance of soil moisture retrievals. This demonstrates the stability of the SMAP_E_OPL processor with respect to different LSMs used to provide ancillary information of T_{eff} to the soil moisture retrieval algorithms. As presented in Table 2, similar standard deviations of the evaluation metrics between the soil moisture retrieval processors are obtained, which indicate that the performance of the soil moisture retrievals is also comparable across the SCAN and USCRN stations within CONUS. Figure 2 further supports this finding, as almost identical spatial patterns of the ubRMSE, RMSE, bias, and R between the soil moisture retrievals from SMAP_E_OPL (JULES), SMAP_L2SMP_E, and SMAP_L2SMP are observed. Overall, the western CONUS exhibits comparatively lower ubRMSE, RMSE, and bias accompanied by higher R values, in contrast to the eastern CONUS. This difference can be attributed to relatively drier conditions (low soil moisture dynamics) in the western CONUS and wetter soil conditions (high soil moisture dynamics) in the eastern CONUS.

The geospatial distribution of evaluation metrics (ubRMSE, RMSE, bias, and R) in Figure 2a corresponds well with the distribution of land cover types (Figure 2m). Specifically, the error metrics (i.e., ubRMSE, RMSE, and bias) tend to be relatively low in shrublands and grasslands, while the R is relatively high in croplands. These patterns are clearly seen in the distribution plots of the metrics (shown in Figure 3) for grasslands, croplands, and shrublands, which are three dominant land cover types within the SCAN and USCRN stations. Figure 3 also shows that the error metrics are more spatially distributed in croplands than in other land cover types, while the R values are more spatially distributed in grasslands and shrublands. Likewise, the level and distribution of the evaluation metrics for different land cover types are comparable between the SMAP_E_OPL and official SMAP soil moisture products.

Note that while the station-to-pixel evaluation approach is insufficient to draw robust conclusions, it is also the approach adopted in the SMAP assessment report (O'Neill et al., 2021b). To further support our results, we present T_B comparisons between the IDSW and BG interpolation methods in Supplementary Text 1 (Supplementary Figure 1; Supplementary Table 3) and soil moisture comparisons between SMAP_E_OPL and SMAP_L2SMP_E over the global domain in Supplementary Text 2 (Supplementary Figures 2-4).

3.2 Enhancing hydrological models: use of SMAP_E_OPL for real-time soil moisture assimilation

The assimilation of soil moisture data obtained through remote sensing with land surface models has been shown to improve the characterization of land surface conditions (Azimi et al., 2020; Baugh et al., 2020; Bolten et al., 2010). In this section, we examine the utility of the real-time SMAP_E_OPL data for data assimilation (DA). Details of the soil moisture DA method are provided in Supplementary Text 3 and Supplementary Table 4, and the experimental setup is described in Supplementary Text 4 and Supplementary Table 5. Soil moisture estimates from the model without assimilation (open loop; OL) and the model integration with DA are evaluated using *in situ* soil moisture measurements from the SCAN and USCRN stations using ubRMSE and R. Spatial mean and standard deviation of the evaluation metrics, percentage improvement in the soil moisture ubRMSE and R by assimilating the SMAP retrievals as compared to the OL soil moisture estimates, and percentage of grid cells improved by DA are presented in Table 3. As shown in Table 3, the domain-averaged ubRMSE and R of the OL soil moisture are 0.052 $m^3\ m^{-3}$ ($\pm$0.019 $m^3\ m^{-3}$) and 0.671 ($\pm$0.117), respectively, and assimilation of the SMAP soil moisture retrievals reduces the ubRMSE and increases the R. The percentage improvement achieved by the SMAP DA ranges from 3.5% to 4.8% for ubRMSE and from 1.8% to 3.4% for R. In the DA integration, soil moisture estimates are enhanced for more than 64% and 58% of model grid cells with respect to the ubRMSE and R, respectively.

It can be noted from Table 3 that the effectiveness of SMAP_E_OPL in improving the soil moisture estimates within the DA system is comparable to that of the official SMAP products (SMAP_L2SMP_E and SMAP_L2SMP) when the results are compared against the SCAN and USCRN measurements. Regarding the domain-averaged ubRMSE, the 9-km products (SMAP_E_OPL and SMAP_L2SMP_E) show slightly better performance than the 36-km SMAP_L2SMP data while SMAP_E_OPL exhibits marginally lower R than that of SMAP_L2SMP_E and SMAP_L2SMP. For both evaluation metrics, the NRT 9-km soil moisture DA (DA_{OPL}) exhibits a slightly lower performance compared to the high-latency 9-km soil moisture DA (DA_{L2E}). However, given the short latency (~3 min when the SMAP NRT L1B T_B is available) of the SMAP_E_OPL data, these minor variations in performance may be inconsequential concerning the practicality of integrating the SMAP_E_OPL into operational soil moisture DA systems.

The spatially distributed differences between the soil moisture ubRMSE and R in DA and OL cases (DA minus OL) are shown in Figure 4. The figure shows that the performance of the SMAP_E_OPL soil moisture DA has a similar spatial pattern to that of the SMAP_L2SMP_E DA. The effectiveness of assimilating all SMAP products to improve the model estimates of soil moisture varies based on dominant land cover types (such as grasslands, croplands, and shrublands) present in the SCAN and USCRN stations (Figure 5). Assimilating SMAP soil moisture data decreases ubRMSE across all three land cover types, and enhances R for grasslands and croplands, while it decreases R for shrublands (Figure 5b). These findings are in agreement with those found from Figure 3 where all three SMAP products (9-km SMAP_E_OPL, 9-km SMAP_L2SMP_E, and 36-km SMAP_L2SMP) exhibited the smallest ubRMSE, RMSE, and bias in all land cover types, but the lowest R in shrublands.

It can be noted from Figures 5c,d that the number of improved grid cells in the SMAP_E_OPL DA is comparable to that found in the official SMAP DA across various land cover types. These findings affirm the feasibility of incorporating the NRT 9-km SMAP soil moisture retrievals (SMAP_E_OPL) into operational data assimilation (DA) systems.

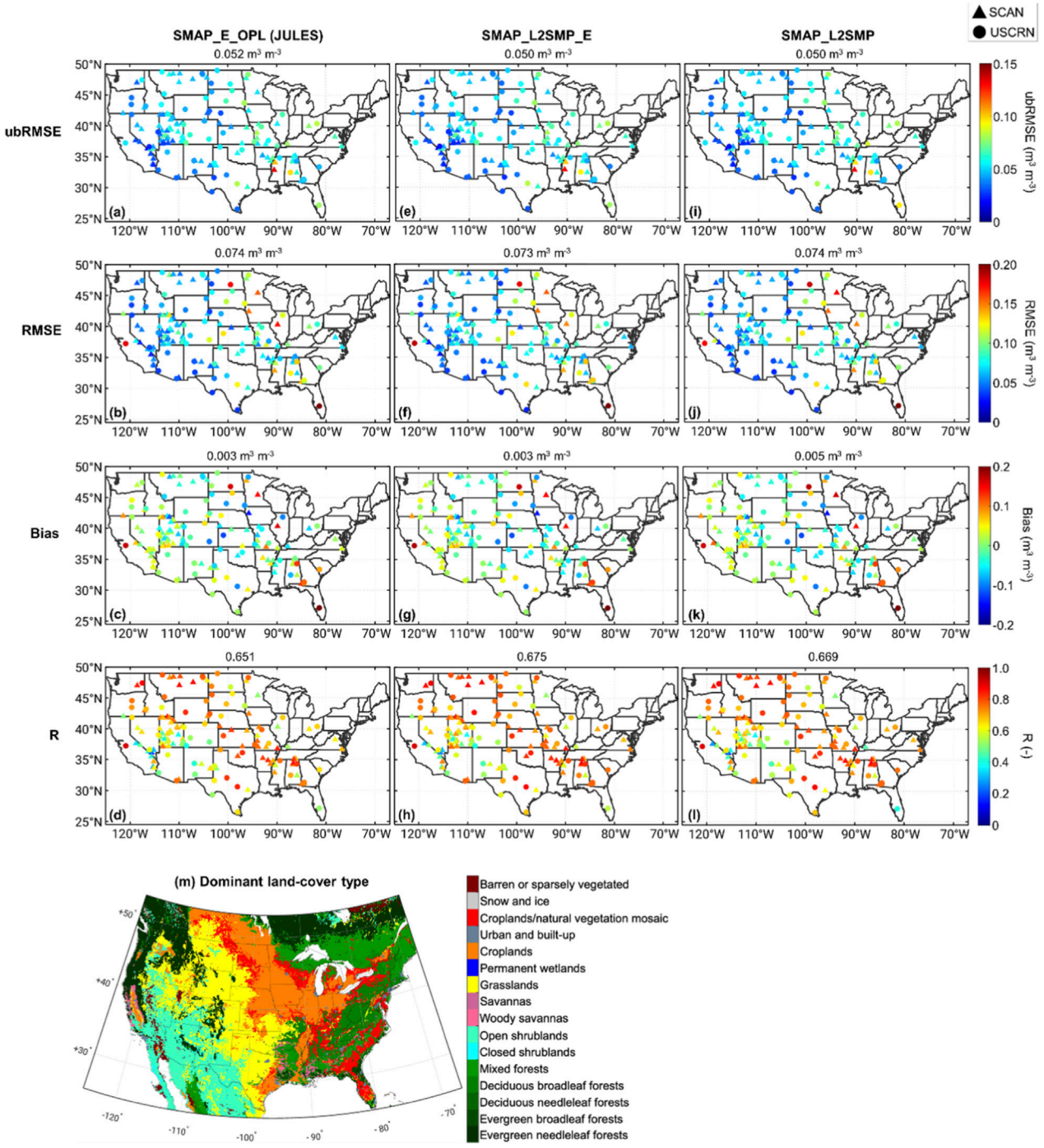


FIGURE 2
Geospatial distribution of the evaluation metrics (ubRMSE, RMSE, bias, and R) of soil moisture retrievals from SMAP_E_OPL using JULES **(a-d)**, SMAP_L2SMP_E **(e-h)**, and SMAP_L2SMP **(i-l)** in the CONUS domain. The metrics are calculated using the *in-situ* soil moisture measurements from the SCAN and USCRN stations during 1 April 2015 to 31 December 2021. Domain-averaged values are also presented above each map. For comparison purposes, a map of dominant land cover types (m) generated from the MODIS-based IGBP land classification (Friedl et al., 2002) is also provided.

4 Discussion: implications for time-sensitive applications

The study implemented the SMAP_E_OPL framework to retrieve soil moisture in NRT, which is then assimilated in LISF with three different LSMs. Results demonstrate that the SMAP_E_OPL provides similar quality of soil moisture retrievals as compared to SMAP L2SMP_E product. The soil moisture estimates from LSMs are also improved after assimilating the NRT soil moisture. The timely, accurate soil moisture provides crucial information

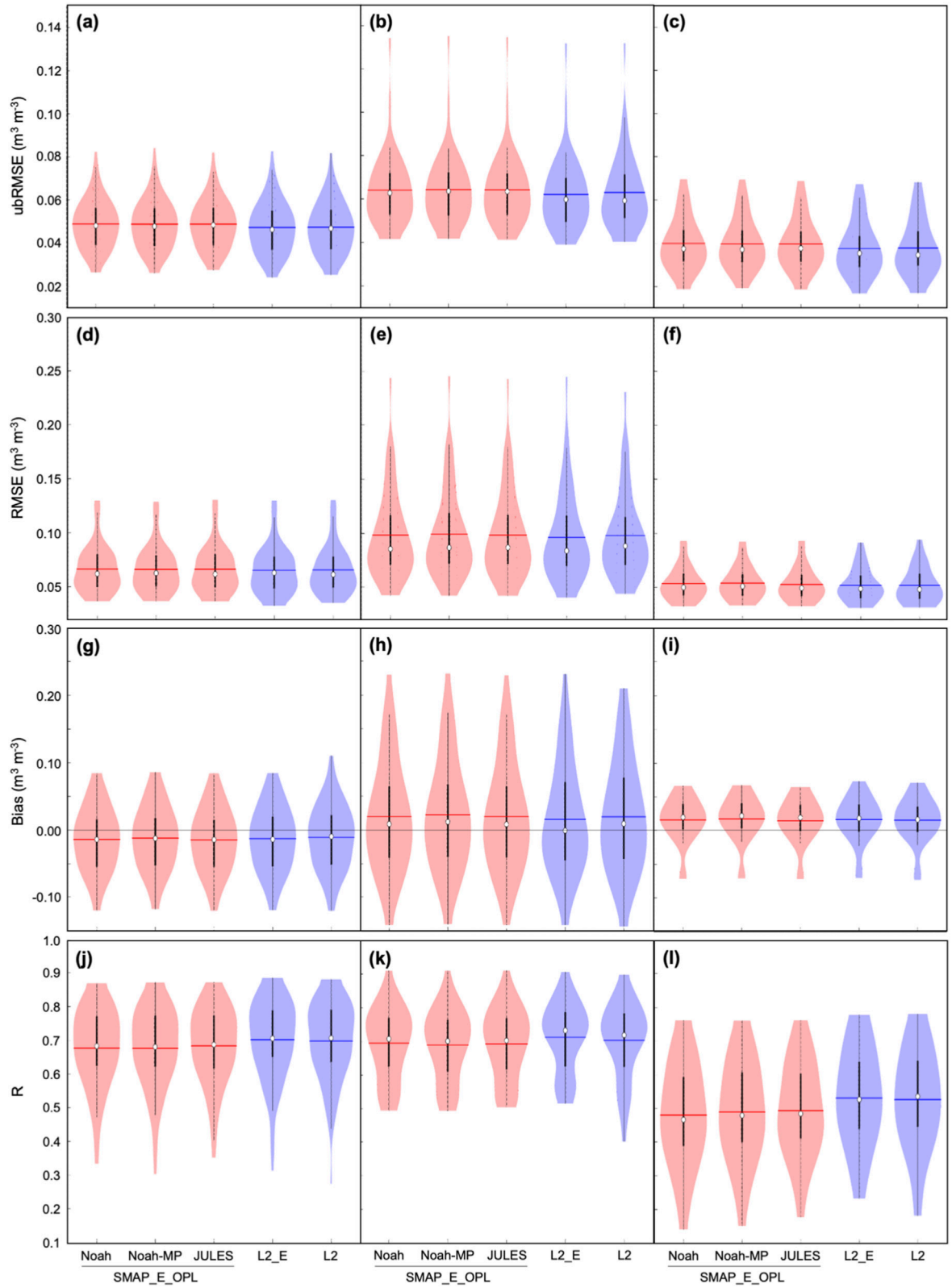


FIGURE 3
Distribution plots of the soil moisture retrieval evaluation metrics (ubRMSE, RMSE, bias, and R) for three dominant land cover types (i.e., grassland, cropland, and shrubland) in the *in-situ* soil moisture measurement stations used in this study. In the distribution plots, the mean and median of the metrics are represented by solid horizontal lines and white dots, respectively, and the 95% confidence interval and interquartile range are indicated by vertical lines and black boxes, respectively. The red and blue shades present sample distribution for the statistics. **(a)** ubRMSE: Grassland. **(b)** ubRMSE: Cropland. **(c)** ubRMSE: Shrubland. **(d)** RMSE: Grassland. **(e)** RMSE: Cropland. **(f)** RMSE: Shrubland. **(g)** Bias: Grassland. **(h)** Bias: Cropland. **(i)** Bias: Shrubland. **(j)** R: Grassland. **(k)** R: Cropland. **(l)** R: Shrubland.

TABLE 3 Summary of the simulated soil moisture evaluation results over CONUS.

Case name	ubRMSE (m^3 m^{-3})			R		
	Mean (± std dev)	Percentage improvement (%)	Improved grid cells (%)	Mean (± std dev)	Percentage improvement (%)	Improved grid cells (%)
OL	0.052 (±0.019)	-	–	0.671 (±0.117)	-	–
DA_{OPL}	0.050 (±0.018)	3.9	73.2 (23.5*)	0.690 (±0.105)	2.9	71.2 (22.2*)
DA_{L2E}	0.050 (±0.019)	3.9	78.4 (19.0*)	0.694 (±0.104)	3.4	75.2 (17.0*)
DA_{L2}	0.050 (±0.018)	3.5	74.5 (15.7*)	0.691 (±0.107)	3.1	71.9 (12.4*)

The evaluation metrics (ubRMSE and R) are calculated by comparing the soil moisture estimates from the OL and DA cases against the SCAN and USCRN soil moisture measurements. Spatial mean (± standard deviation) of ubRMSE and R, percentage improvement by DA as compared to the OL run, and percentage of improved grid cells are presented.
*Percentage of statistically significantly ($p < 0.05$) improved grid cells (p-values are obtained by t-test).

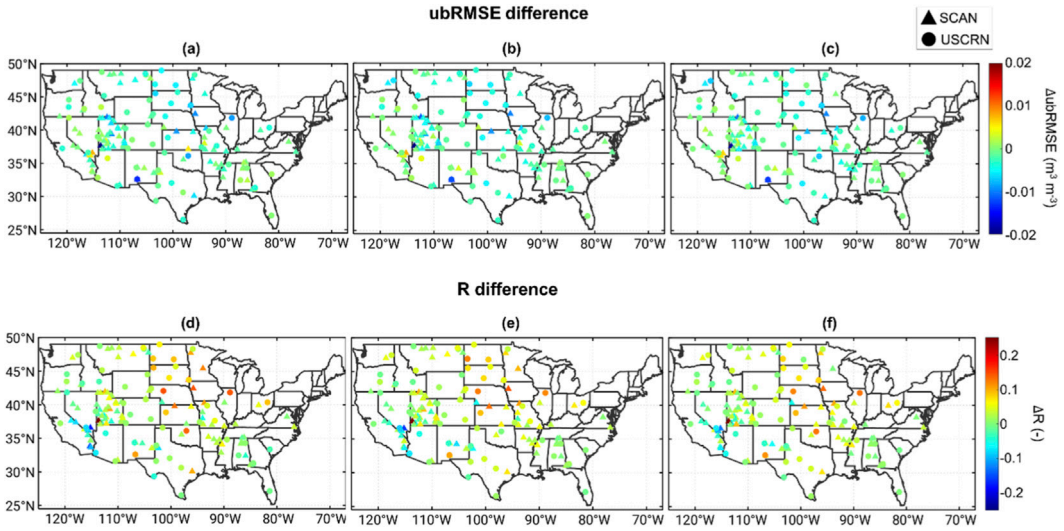


FIGURE 4
Surface layer soil moisture ubRMSE and R differences between the DA and OL cases ($\Delta ubRMSE = ubRMSE_{DA} - ubRMSE_{OL}$ and $\Delta R = R_{DA} - R_{OL}$, respectively) for the SCAN and USCRN stations within CONUS. Negative and positive values indicate that the model estimates of soil moisture are improved by assimilating the SMAP soil moisture retrievals with respect to the ubRMSE and R, respectively. **(a)** DA_{OPL}-OL. **(b)** DA_{L2E}-OL. **(c)** DA_{L2}-OL. **(d)** DA_{OPL}-OL. **(e)** DA_{L2E}-OL. **(f)** DA_{L2}-OL.

for various applications. For example, soil moisture and rainfall are the two key factors in flash flood development and landslides in mountainous areas (Mao et al., 2019). About 60% of rainfall over land originates from land evaporation, which is driven by the soil-plant-atmosphere interaction (Brocca et al., 2017). NRT soil moisture data, ideally within 3 h from time of acquisition (Brocca et al., 2017; Rodríguez-Fernández et al., 2019), enhances the initialization of land surface conditions for numerical weather forecasting and produces more accurate estimates of the rainfall, air temperature and humidity near the surface (Deardorff, 1977; Koster et al., 2004; Kwon et al., 2024; 2026). These estimates from the timely forecasting system provide valuable information for early-warning and emergency disaster response systems. Incorporating antecedent soil moisture conditions with forecasted rainfall can improve flood peak prediction (Baugh et al., 2020; Grillakis et al., 2016) and thus provide timely warning information for disaster mitigation and emergency response to upcoming flash floods and landslides (Biswas et al., 2022; Zhao et al., 2019). In this respect, although it still has room for further improvement, SMAP_E_OPL can potentially provide great benefits for managing and mitigating natural disaster-induced damages by delivering timely (within 3 h) high-spatial-resolution soil moisture data.

5 Summary and conclusions

This study implements the operational enhanced SMAP soil moisture retrieval processor (i.e., SMAP_E_OPL) in the NASA LISF with the aim of producing NRT hourly soil moisture products at 9-km spatial resolution for use within operational systems. To achieve this, the standard retrieval procedure is followed, using the same inversion algorithms and preprocessed ancillary data as used for the official SMAP_L2SMP_E soil moisture products.

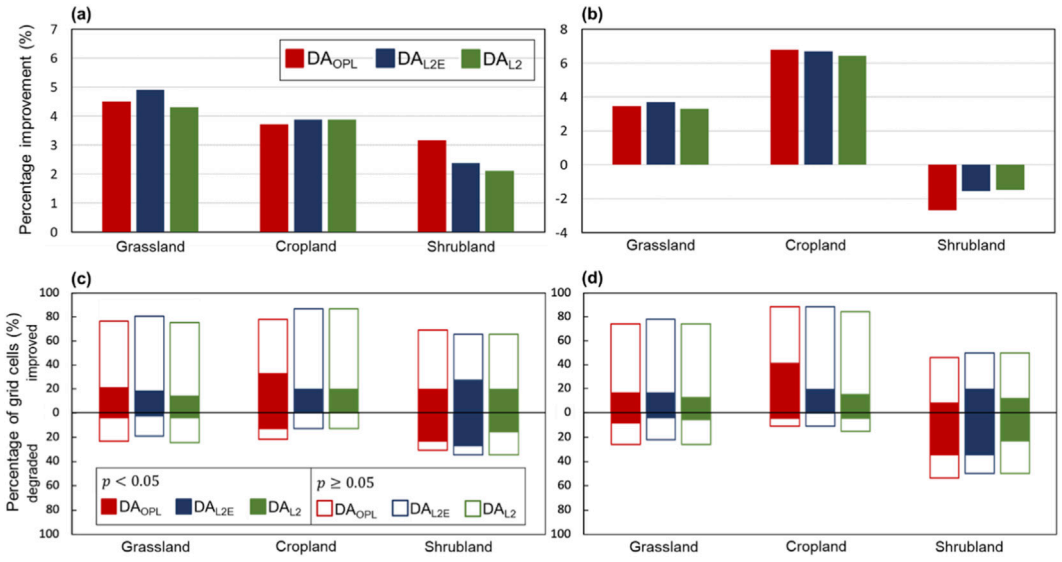


FIGURE 5
Percentage improvement in soil moisture estimates by DA as compared to the OL run **(a,b)**, and the percentage of grid cells whose soil moisture estimates are improved or degraded by DA **(c,d)** with respect to ubRMSE **(a,c)** and R **(b,d)**. The evaluation results are presented for three dominant land cover types (grasslands, croplands, and shrublands) in SCAN and USCRN stations. *p*-values are obtained by *t*-test. **(a)** Percentage improvement in ubRMSE, **(b)** Percentage improvement in R, **(c)** Percentage of improved and degraded grid cells for ubRMSE, and **(d)** Percentage of improved and degraded grid cells for R.

However, efforts have been made to reduce the reliance of SMAP_E_OPL on data sources with high temporal latency. Specifically, SMAP_E_OPL relies solely on the SMAP L1B T_B data and uses soil temperature and snow depth outputs from the NRT US Air Force 557th WW global operational LIS-LSMs such as Noah, Noah-MP, and JULES. Furthermore, to enhance processing efficiency, SMAP_E_OPL adopts the IDSW interpolation method, instead of the BG method used in SMAP_L2SMP_E. This modification serves to decrease the processing time while upholding the integrity of the outcomes. Additionally, bias-correction of the LIS-LSM-based effective temperature (T_{eff}) is performed to ensure consistency with the SMAP_L2SMP_E product.

The SMAP_E_OPL soil moisture is produced over a global domain at a spatial resolution of 0.14° (longitude) by 0.09° (latitude). In this study, we evaluate the performance of the SMAP_E_OPL product over CONUS using *in situ* soil moisture measurements from SCAN and USCRN. The performance of the SMAP_E_OPL soil moisture retrievals is found to be comparable to that of the official (high latency) SMAP soil moisture products (9-km SMAP_L2SMP_E and 36-km SMAP_L2SMP) based on domain-averaged evaluation metrics. SMAP products (SMAP_E_OPL, SMAP_L2SMP_E and SMAP_L2SMP) exhibit similar spatial patterns of performance, which correspond to those of land cover types. Using different LSMs (Noah, Noah-MP, and JULES) to estimate T_{eff}, SMAP_E_OPL achieves equivalent performance in retrieving soil moisture, which implies flexibility of the SMAP_E_OPL processor in terms of LSMs.

In conclusion, this study demonstrates that SMAP_E_OPL achieves a level of performance comparable to that of the official SMAP soil moisture products, while providing high-spatial-resolution soil moisture data with significantly reduced latency. This implies the feasibility of SMAP_E_OPL to be used in NRT operational forecasting and monitoring systems. Particular emphasis is placed on the advantages of SMAP_E_OPL relative to existing NRT SMAP products in terms of (a) enhanced posted grid resolution compared to the standard 36-km NRT product, (b) generation of NRT hourly soil moisture data while preserving the original 2–3 days revisit time and without relying on external data sources [compared to 1-km NRT daily products (e.g., Yin et al., 2020)], and (c) the flexibility of the SMAP_E_OPL processor for application in various operational systems. Thus, the development of the SMAP_E_OPL soil moisture retrievals is a promising capability for improving land characterization within operational environments.

However, the following limitations should be addressed in future studies before the operational implementation of the SMAP_E_OPL processor at the global scale. First, the evaluations in this study are based on station-to-pixel comparisons following the Sparse Network Assessment used in the official SMAP quality assessment report (O'Neill et al., 2021b), along with the quality-control protocols of Chen et al. (2018). We acknowledge that this approach is subject to representativeness error. In addition, ground-measurement-based evaluations are conducted over CONUS to ensure the quality of soil moisture estimates, whereas global-scale evaluations are performed based on comparisons between SMAP_E_OPL and SMAP_L2SMP_E, using the latter as a reference (Supplementary Text 2 and Supplementary Figures 2-4), because the primary objective of this study is to develop SMAP_E_OPL with performance comparable to that of SMAP_L2SMP_E. Future studies may further evaluate these SMAP soil moisture products using more spatially distributed metrics, such as triple collocation analysis, while also considering

periods of extreme hydrometeorological events, to better support the operational applicability of the developed processor. Second, this study presents the overall performance of SMAP_E_OPL after applying all major design differences relative to SMAP_L2SMP_E, including the interpolation method, spatial grid/projection, and sources of effective soil temperature simulation and snow/frozen-soil flags (Table 1). To improve future modifications of SMAP_E_OPL, explicit analyses of the impact of each design change on soil moisture retrieval performance are suggested. Third, as noted, the current version of the SMAP_E_OPL processor employs SCA-V for soil moisture retrieval, whereas DCA has recently been adopted as the SMAP baseline algorithm. Future versions of the SMAP_E_OPL processor may benefit from using DCA, along with evaluations over a more recent data period. Lastly, the IDSW interpolation method may have limitations in regions with strong land surface heterogeneity because it does not explicitly account for data-to-data correlation and can be sensitive to irregularly distributed data (Li et al., 2020; Liu P.-W. et al., 2021; Lu and Wong, 2008), although it showed interpolation performance comparable to that of the BG method in this study (Supplementary Text 1, Supplementary Figure 1; Supplementary Table 3). More sophisticated yet computationally efficient interpolation methods should be explored in future studies.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

YK: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. P-WL: Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – review and editing. MN: Investigation, Validation, Visualization, Writing – review and editing. RB: Conceptualization, Investigation, Methodology, Supervision, Writing – review and editing. EK: Investigation, Software, Writing – review and editing. JW: Project administration, Writing – review and editing. EJ: Investigation, Writing – review and editing. SK: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing – review and editing.

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Conflict of interest

Authors P-WL, EK, JW, and EJ were employed by Science Systems and Applications Inc.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/feart.2026.1854083/full#supplementary-material

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Glossary

AMSR-E	Advanced Microwave Scanning Radiometer for Earth Observing System
ATBD	Algorithm theoretical basis document
BATS	Biosphere-atmosphere transfer scheme
BG	Backus-Gilbert
CONUS	Continental United States
DA	Data assimilation
DCA	Dual-channel algorithm
EASE	Equal-area scalable earth
FLDAS	Famine early warning land data assimilation system
GALWEM	Global air-land weather exploitation model
GEOS-FP	Goddard earth observing system-forward processing
GFS	Global forecast system
GLDAS	Global land data assimilation system
GMAO	Global modeling and assimilation office
GRMDM	Generalized refractive mixing dielectric model
IDSW	Inverse distance squared weighted
IGBP	International geosphere-biosphere programme
IMERG	Integrated multiscale retrievals for global precipitation measurement
IMS	Interactive multisensor snow and ice mapping system
JULES	Joint UK land environment simulator
LDT	Land data toolkit
LIS	Land information system
LISF	Land information system framework
LSM	Land surface model
L1C_TB_E	Level 1C enhanced brightness temperature product
MODIS	Moderate resolution imaging spectroradiometer
NASA	National aeronautics and space administration
NDVI	Normalized difference vegetation index
NLDAS	North American land data assimilation system
NOAA	National oceanic and atmospheric administration
Noah-MP	Noah LSM with multi-parameterization options
NRT	Near-real-time
NWP	Numerical weather prediction
OL	Open loop
OSU	Oregon state university
R	Correlation coefficient
RMSE	Root mean square error
SCAN	Soil climate analysis network
SCA-H	Single channel algorithm using horizontally polarized brightness temperature
SCA-V	Single channel algorithm using vertically polarized brightness temperature
SIMGM	Simple groundwater scheme
SM	Soil moisture
SMAP	Soil moisture active passive
SMAP_E_OPL	Operational, enhanced SMAP soil moisture retrieval
SMAP_L2SMP	SMAP level 2 standard soil moisture retrieval
SMAP_L2SMP_E	SMAP level 2 enhanced soil moisture retrieval
T_B	Brightness temperature
T_{eff}	Effective soil temperature
TOPMODEL	TOPography-based hydrological MODEL
t-1 day	Same time of the previous day
***t*-3h**	3 h before
ubRMSE	Unbiased root mean square error
UK	United Kingdom
US	United States
USAF	United States air force
USAF-SI	United States air force snow and ice analysis
USCRN	United States climate reference network
USDA	United States department of agriculture
VIIRS	Visible infrared imaging radiometer suite
VWC	Vegetation water content
τ	Vegetation optical depth
ω	Albedo scattering
557th WW (557WW)	557th weather wing